Tick, Dollar and Volume Bars: Smarter Sampling of Financial Data (with code)

By QuantBeckman

A detailed, code-first newsletter piece on turning a raw high-frequency stream of timestamped trades into model-ready bars, framed around 'what are your bars hiding from you?'. The analogy: fixed-time sampling is a security camera photographing every minute regardless of activity; event-based sampling is a motion-activated camera that records only when something significant happens — a certain volume, price move, or dollar value.

It argues the sampling decision is the first and most profound obstacle in a quant pipeline, made before any strategy design, and works through time bars, tick bars, volume bars, VWAP bars and dollar bars with Python code. Explicit about each method's limitations and the engineering reality that raw tick data is granular but computationally heavy and statistically awkward, so bar construction compresses the (t, p, v) stream into representations with better statistical properties.

Note: a paid/premium newsletter post — the full code and later sections sit behind a subscription. Archived as an original summary card.

Source: https://www.quantbeckman.com/p/what-are-your-bars-hiding-from-you